

Empowering Parents and Teachers to Support Children's Learning through AI-based and Robotic Learning Companions

Hui-Ru Ho

Department of Computer Sciences
University of Wisconsin-Madison
Madison, Wisconsin, USA
hho24@cs.wisc.edu

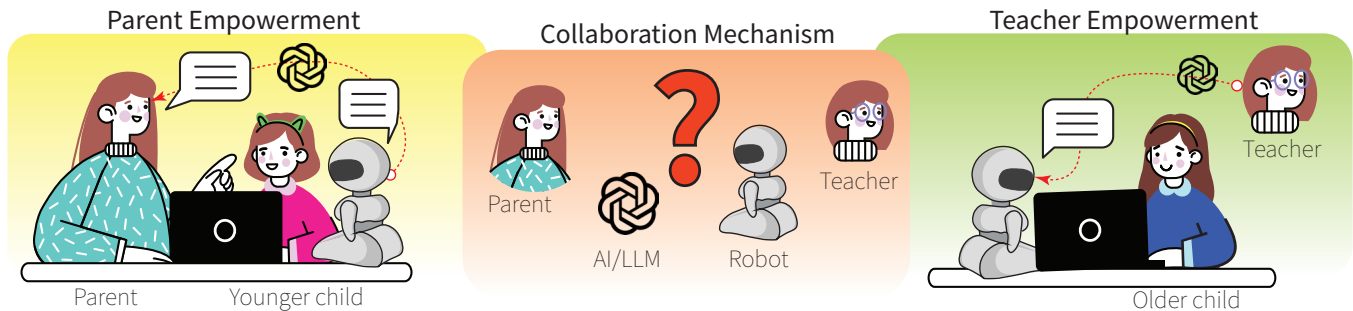


Figure 1: My work focus on empowering parents and teachers to support children's learning through AI-based and robotic collaborative systems.

Abstract

Human-centered AI and robots can augment the capacity of teachers and parents in educating children through collaboration, empowering them to overcome limitations such as skill levels, energy and time to deliver higher quality of interaction with children. My current research focuses on understanding and designing technologies enabled by AI-assisted social robots to support parenting practices to enhance informal in-home learning of young children. In addition, I developed comprehensive framework and concrete prototypes to demonstrate and evaluate novel interaction paradigms that enable seamless collaboration between parents and robots to support children's learning at home. My next step involves expanding my research to encompass both formal and informal learning settings, using a human-centered approach to iteratively design, implement, and evaluate novel human-AI collaboration mechanisms for educational robots. The goal is to empower teachers and parents, ultimately improving children's learning outcomes and experiences in a way that is ethical, responsible, and developmentally-appropriate.

CCS Concepts

• **Human-centered computing** → **Collaborative interaction**;
Empirical studies in HCI; **Empirical studies in interaction**

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3707605>

design; • **Applied computing** → **Interactive learning environments**.

Keywords

Human-robot interaction, human-AI interaction, family and education, user-centered design

ACM Reference Format:

Hui-Ru Ho. 2025. Empowering Parents and Teachers to Support Children's Learning through AI-based and Robotic Learning Companions. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3706599.3707605>

1 Research Motivation

As children develop through different stages, teachers and parents play pivotal roles in their learning ecosystems [2, 7]. However, the level and quality of support often vary due to factors such as student-teacher ratios [10], parental education levels [12], and parent availability [11, 18]. These disparities in access to resources provided by teachers and parents contribute to inequality and discrepancies in children's development. Advancements in AI and robots, particularly in their intelligence and autonomy, hold promising potential to promote equity in access to high-quality learning assistance [27, 37]. Therefore, it is crucial to explore how we can leverage these technologies to empower key educators—parents and teachers—thereby enhancing children's learning experiences, outcomes, and opportunities.

Despite the development of educational technologies that provide personalized and adaptive learning support for children [41], many similar systems focus on individual tutoring rather than empowering the key stakeholders responsible for education. These systems were designed for “child-centered” needs and may require

children to be mature or independent enough to use, as they were not designed considering the involvement of teachers and parents. However, problems related to unsupervised use of technology have emerged dramatically for children, raising debates about safety and ethics of designing technology for younger populations [14, 15, 37]. People expressed concerns about the use of AI and robots in education, including privacy [3], content appropriateness [15], and misinformation [22]. For example, children may not be capable of making judgments on falsified information and may rely on tools like generative AI to write essays and homework. In addition, worries related to the absence of human-human connections arise as most educational technology focus on child-centered design and needs. Many studies discussed the roles of advancing AI and robots, debating whether they will replace parents and teachers [8, 18]. These issues further emphasize the importance of considering flexible involvement of educators in mediating technology use for children.

Moreover, children's development results in dynamic needs of support from both parents and teachers over time. For example, parents are especially crucial in fostering intellectual and social-emotional skills in younger children before they enter formal schooling [16, 30]. Yet, as children mature and begin learning complex concepts, the role of teachers becomes more prominent, as advanced pedagogical skills are required for effective instruction. The design of, for example, an in-home robot may shift from handling parent-child-robot interactions to focusing on child-robot interaction using teachers' input. Such design concept considering the developmental changes of children is gradually feasible as AI and robotic techniques evolve. Therefore, my work aims to advance interaction design for human-AI and human-robot collaboration, empowering teachers and parents to support children's age-appropriate learning.

2 Related Work

2.1 Human-centered AI for Education

AI tools can personalize learning by answering questions and tailoring educational content [15, 23, 32, 37, 44]. Research shows AI technologies like ChatGPT support language learning and provide personalized and timely feedback [15, 32], and that children aged 8–14 years report positive learning experiences when interacting with AI-generated content in schools [23]. In addition, AI-driven conversational systems can teach children mathematical concepts through co-creative storytelling, achieving learning outcomes similar to human-led instructions [44].

Moreover, AI can support parents by providing evidence-based parenting advice, *e.g.*, teaching parents how to praise their children [6, 9], providing parenting education assistance [34]. Rather than giving children direct access to AI tools, some parents prefer to use such tools to create educational materials for their children [14]. In addition to learning materials, platforms like ChatGPT offer resources to help parents learn how to teach complex concepts, improving parent-child interactions and addressing time and energy constraints [32, 37, 38]. In addition to parents, teachers also benefit from collaboration with AI to alleviate their workload on repetitive and time-consuming tasks such as grading and taking attendance [20, 21]. In addition, AI can support teachers in making better decisions in instructional design by providing student data analytics

[33] and generating adaptable learning materials through ideation enhancement [14, 15]. Furthermore, prior work also demonstrated how teachers can collaborate with AI in real time to co-orchestrate classroom by supporting student dynamic transitions [42].

Despite these benefits, several concerns persist regarding the use of AI in education. Studies have highlighted the limitations of AI, such as its limited effectiveness in more complex learning tasks, the limited quality of the training data, and its inability to offer comprehensive educational support [37]. There is also a significant risk of AI producing inaccurate or biased information, discouraging independent thought among children, and threatening user privacy [14, 15, 37, 38]. Many parents are skeptical about the role of AI in their children's academic processes, concerned about the accuracy of AI-generated content, and worry that over-reliance on AI could stifle independent thinking [14]. Teachers have also voiced concerns about student agency and the difficulty of distinguishing between AI-generated and student-generated work, further complicating the educational landscape [15].

2.2 Social Robots for Education

Social robots are an emerging technology capable of offering human-like, physically embodied interactions, adding a social dimension that can enhance individual learning experiences [25, 31]. While they offer adaptive assistance and verbal interaction similar to virtual agents [35, 36], their physical presence encourages deeper engagement with the environment and facilitates more advanced social behaviors during learning [1], ultimately improving learning outcomes [27]. Prior work has demonstrated that social robots can effectively facilitate both school-based learning, such as literacy [24] and math [28], as well as home-based learning activities, including reading [31], board games [17], and math-related conversations with parents [19]. Prior work explored how social robots can support children's learning in different interaction settings involving or not involving teachers or parents. First, in *robot-single child* interaction, a robot can effectively facilitate children's independent learning. For example, Chen et al. [5] introduced an adaptive role-switching model that allowed children to alternate between being a tutor or a tutee, dynamically adapting to their personalized learning needs in vocabulary lessons. While Kennedy et al. [24] demonstrated that children could learn elements of a second language from a robot in short-term interactions, Tanaka and Kimura [39] found that children who took on the role of teaching the robot gained confidence and improved learning outcomes.

Second, in *parent-child-robot* interactions, some studies revealed that parents generally feel positive toward incorporating robots into their children's learning, yet they have concerns about technical challenges and replacement of human roles [26, 29, 40]. Gvirsman and Gordon [13] found that, in a parent-toddler-robot interaction, parents tend to decrease their scaffolding affectively when the robot increases its scaffolding behavior. In addition, Chen et al. [4] found that social robots enhanced parent-child co-reading activities and recommended that future designs take into account the preferences of individuals within the child-robot-adult group in collaborative settings. Finally, in a *teacher-children-robot* interaction, typically in a classroom setting, teachers leverage a social robot in a classroom to assist teaching practices and promote student engagement. For

instance, You et al. [43] introduced a humanoid robot into a English class as a teaching assistant to support the teacher motivating a group of children. These studies suggest how social robots could be designed to participate in children’s learning in diverse interaction dynamic contexts, serving various roles such as an individual tutor or a teaching assistant.

3 Research Objectives

Previous research has demonstrated how AI and robots can support children’s learning without adult supervision, addressing limitations in the capabilities, availability, and motivation of teachers and parents. However, concerns about potential risks, ethical issues, and the replacement of human interaction persist. Therefore, it is crucial to design appropriate human-AI collaborations—especially in the form of educational robots—that carefully balance risks and benefits while meeting children’s developmental needs. While a few studies have explored how AI and robots can be used independently by parents and teachers to support learning [5, 32], most have focused on the functional outcomes of these applications (e.g., AI helping teachers create materials, robots assisting parents with tutoring), rather than explicitly addressing the design of the *collaboration mechanisms* between AI, robots, and humans, and how these mechanisms should be responsibly implemented in real-world scenarios to empower teachers and parents.

My research explores how human-AI collaboration—in the form of a learning companion robot—can be designed and developed to support children’s learning by empowering parents and teachers. I believe that the human interactions within parent-child and teacher-student relationships hold irreplaceable value in fostering emotional connections and social development, which are essential components of effective learning that technology alone cannot replicate. The goal is to adopt a human-centered perspective in designing and developing collaboration mechanisms for AI-based and robotic tools while ensuring a balanced approach to human agency, responsibility, and workload for primary educators. I aim to make the following contributions and answer the corresponding research questions:

- (1) (**Understand**) *What are the contexts for empowering teachers and parents to support children’s learning?* – I aim to systematically explore scenarios and contexts where teachers and parents face limitations in supporting children, drawing on frameworks from previous family and education studies.
- (2) (**Design**) *What collaboration mechanisms in AI-based and robotic systems can responsibly empower teachers and parents?* – I aim to analyze contexts and scenarios from formative studies and literature reviews to co-design interaction concepts, workflows, and paradigms with teachers, parents, and children.
- (3) (**Development**) *What AI behaviors and robot interactions are needed to operationalize the collaboration paradigm?* – I aim to develop AI-based robotic systems that demonstrate the implementation of the proposed collaboration mechanisms during the design phase.

- (4) (**Evaluation**) *How effective are the AI-based robotic artifacts in collaborating with teachers and parents in real world settings?* – I aim to evaluate how effective these artifacts empower teachers and parents to support children’s learning through field deployment studies.

4 Contributions to Date

4.1 Design for Informal Learning with Parents and Young Children

Exploring Design of In-Home Educational Robot for Young Children [17, 19]. In my prior work, I focused on understanding the design space and exploring design guidelines for an educational robot that can support young children with learning activities at home. First, I conducted an exploratory study by designing a prototype robot that guided young children to play a number board game with their parents over four days. The qualitative results show that while the robot’s capabilities enhanced children’s learning experiences during the game, the children faced cognitive and emotional challenges such as misalignment of skill level with the game, frustration, and boredom. We recommended that the robot deliver verbal and social cues as feedback, use adaptive scaffolding strategies, and introduce variations in the game to maintain engagement [17]. More importantly, I noticed the inevitability of parental involvement in learning activities related to advanced concepts for young children, leading me to investigate how we can expand existing child-robot interaction paradigms to include parents. Therefore, in another work, I explored how we can design parent-child-robot triadic interactions to facilitate educational conversations, such as *math talk*, a phenomenon where parents incorporate math concepts in daily-life conversations with their children. In the study, parent-child dyads participated in reading sessions with an autonomous robot and free-play activities with a Wizard-of-Oz (WoZ) robot, during which the robot prompted parents to engage in math talk. The qualitative findings offer insights into how parents were inspired by the robot’s prompts, their envision of sharing responsibility with the robot, and how they prefer the robot to initiate conversations with mixed autonomy based on real-life contexts [19].

Designing for Parent-Robot Collaboration Mechanism [18]. My first two projects inspired me to investigate how educational robots can be designed to support multiparty, parent-child dyads interaction, as I realize how important it is for parents to be involved to support young children to advance in learning. I leveraged insights from Ho et al. [19] and conducted a rigorous investigation into how parents would want to use an educational robot at home to support their young children in learning activities using a technology probe (i.e., a prototype learning companion robot with the ability to deliver educational prompts and responses to parent-child pairs during reading sessions autonomously). The study data revealed that parents wanted to collaborate with the robot to support their child’s learning, introducing the concept of “**parent-robot collaboration**”. I further proposed the CAM framework—capability (C), availability (A), and motivation (M)—to model shared responsibility between parents and robots across various scenarios, guiding robot use and shaping interaction dynamics. In addition, I emphasized

that parents prefer to define the overarching educational goals delivered by the robot and to mitigate their concerns when using the robot. This work contributes a comprehensive framework for how robots can be designed to foster parent-robot collaboration, explaining why and how parents can collaborate with an educational robot to support their young children [18].

Developing Parent-AI Collaboration Platform using an Educational Robot to support Young Children (Under review). In this work, I aim to develop design methods and artifacts to operationalize the CAM framework from Ho et al. [18]. First, I created a card-based activity kit, SET, to systematically map parental involvement scenarios. In addition, my group implemented PAiRED, an LLM-assisted interface integrated with an educational robot, enabling parents to review and revise LLM-generated learning content while adjusting their level of involvement by delegating roles between themselves and the robot. I conducted an in-home study with 20 parent-child pairs (children aged 3–5 years old) to explore parents' involvement scenarios, usage of LLM-generated content, and collaboration patterns with an educational robot in varying contexts. I found that, although parents generally feel confident in their skills, they seek additional guidance in pedagogical strategies, especially for advanced learning concepts. Parents also hold mixed attitudes toward AI-generated content, prioritizing factors such as difficulty, variety, and quality when reviewing it. Furthermore, parents vary in how they use AI-generated content, choosing to edit, skim, reuse, use directly, or even avoid it, and they value the pedagogical insights provided by AI. Finally, I identified collaboration patterns between parents and the robot, emphasizing that parent-robot collaboration should extend beyond task distribution and that content difficulty should adapt to interaction dynamics to maximize the value of parental involvement. This work provides a novel approach to capturing parental involvement scenarios, offers deeper insights into parent interactions with AI-generated learning content, and presents an innovative interaction paradigm for dynamically adjusting involvement based on scenario changes.

4.2 Design for Formal Learning with Middle-School Children

Exploring Design of Learning Companion Robots for Assisting Middle-School Children in Homework (Under review). To expand my work from informal learning settings and younger children to formal learning environments and older students, I conducted an exploratory study on the interaction design of a homework companion robot for middle-school students. I reported findings from in-home technology probe studies and interviews with 10 students (aged 10–12), examining their expectations for the robot's interaction modalities and behaviors. This included desired feedback, communication methods, student perspectives, and medium artifacts in human-robot interaction. Based on these insights, I developed design recommendations for modality personalization, adapting homework formats, and optimizing robot personas.

5 Expected Next Steps

I aim to expand my current work by advancing the novelty of *collaboration mechanisms* with AI-based and robotic agents to ethically,

responsibly, and dynamically empower parents and teachers in supporting children's learning throughout their developmental stages. Below, I outline the anticipated future directions of my research.

Exploring Human-AI Collaboration for Learning Companion Robots in Formal and Informal Learning Settings. Most of my work has focused on designing for young children aged 3–5 and involving their parents. However, previous studies have revealed that many parents envision AI-based and robotic systems that integrate teachers' assessments to support their children's development by guiding which learning activities to prioritize at home. To explore how insights from formal schooling can be translated into in-home technological support, it is essential to expand research to include formal learning settings and older children.

Design and Evaluation of Human-AI Collaboration Mechanisms in Robotic Learning Companions for Parents and Teachers. While I have proposed an initial prototype for a human-AI and human-robot collaboration mechanism tailored to parents, our user studies identified areas for improvement, which are detailed in our paper. My goal is to incorporate these design insights to enhance the mechanism and develop improved system prototypes. Specifically, I plan to integrate mixed-level control mechanisms that allow users to seamlessly co-create learning content with LLMs, refine prompt engineering to improve the quality of auto-generated questions, and adapt interaction modalities to align with children's developmental needs and levels of adult involvement. Additionally, I will conduct field experiments with teachers, parents, and children from diverse demographic backgrounds to evaluate the effectiveness of the system in facilitating collaboration between teachers, parents, and AI-based robotic agents in supporting children's learning.

6 Dissertation Status and Long-term Goals

I am a third-year Ph.D. student in the Computer Sciences Department at the University of Wisconsin-Madison, working under the guidance of Dr. Bilge Mutlu. I completed my required coursework and earned my master's degree in 2023. My goal is to complete my qualifying exam in Spring 2025, finish my preliminary exam by Spring 2026, and defend my dissertation by Spring 2027. After graduation, I aim to become a tenure-track professor, advancing interdisciplinary research in HCI and Education. My academic focus is on designing, developing, and evaluating novel interaction paradigms that enable human-AI and human-robot collaborations to empower educators and caregivers, fostering child development and family well-being. Participation in the CHI'25 Doctoral Consortium will offer invaluable opportunities for feedback, insights, and mentorship from fellow researchers, particularly as I have not yet attended a doctoral consortium at any conference.

Acknowledgments

I am grateful to my academic advisor professor Bilge Mutlu and collaborator professor Edward Hubbard for their continuous mentorship and support. I also thank my colleagues from the People and Robots Lab for their support throughout my academic journey.

References

- [1] Tony Belpaeme, James Kennedy, Aditi Ramachandran, Brian Scassellati, and Fumihide Tanaka. 2018. Social robots for education: A review. *Science robotics* 3,

- 21 (2018), eaat5954.
- [2] Belinda Blevins-Knabe and Ann M Berghout Austin. 2016. *Early childhood mathematics skill development in the home environment*. Springer.
- [3] Bengisu Cagiltay, Hui-Ru Ho, Joseph E Michaelis, and Bilge Mutlu. 2020. Investigating family perceptions and design preferences for an in-home robot. In *Proceedings of the interaction design and children conference*. 229–242.
- [4] Huili Chen, Anastasia K Ostrowski, Soo Jung Jang, Cynthia Breazeal, and Hae Won Park. 2022. Designing Long-term Parent-child-robot Triadic Interaction at Home through Lived Technology Experiences and Interviews. In *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 401–408.
- [5] Huili Chen, Hae Won Park, and Cynthia Breazeal. 2020. Teaching and learning with children: Impact of reciprocal peer learning with a social robot on children's learning and emotive engagement. *Computers & Education* 150 (2020), 103836.
- [6] Deborah Winders Davis, M Cynthia Logsdon, Krista Vogt, Jeff Rushton, John Myers, Adrian Lauf, and Felicia Hogan. 2017. Parent education is changing: a review of smartphone apps. *MCN: The American Journal of Maternal/Child Nursing* 42, 5 (2017), 248–256.
- [7] Greg J Duncan, Chantelle J Dowsett, Amy Claessens, Katherine Magnuson, Aletha C Huston, Pamela Klebanov, Linda S Pagani, Leon Feinstein, Mimi Engel, Jeanne Brooks-Gunn, et al. 2007. School readiness and later achievement. *Developmental psychology* 43, 6 (2007), 1428.
- [8] Bosede I Edwards and Adrian D Cheok. 2018. Why not robot teachers: artificial intelligence for addressing teacher shortage. *Applied Artificial Intelligence* 32, 4 (2018), 345–360.
- [9] Guido A Entenberg, Gilly Dosovitsky, Shirin Aghakhani, Karin Mostovoy, N Carre, Zendrea Marshall, D Benfica, S Mizrahi, A Testerman, A Rousseau, et al. 2023. User experience with a parenting chatbot micro intervention. *Frontiers in Digital Health* 4 (2023), 989022.
- [10] Alicia K Green. 2024. Examining the Factors Contributing to the Teacher Shortage Crisis in the US. (2024).
- [11] Christa L Green, Joan MT Walker, Kathleen V Hoover-Dempsey, and Howard M Sandler. 2007. Parents' motivations for involvement in children's education: An empirical test of a theoretical model of parental involvement. *Journal of educational psychology* 99, 3 (2007), 532.
- [12] Wendy S Grolnick, Corina Benjet, Carolyn O Kurowski, and Nicholas H Apostoleris. 1997. Predictors of parent involvement in children's schooling. *Journal of educational psychology* 89, 3 (1997), 538.
- [13] Omer Gvirsman and Goren Gordon. 2024. Effect of Social Robot's Role and Behavior on Parent-Toddler Interaction. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*. 222–230.
- [14] Ariel Han and Zhenyao Cai. 2023. Design implications of generative AI systems for visual storytelling for young learners. In *Proceedings of the 22nd Annual ACM Interaction Design and Children Conference*. 470–474.
- [15] Ariel Han, Xiaofei Zhou, Zhenyao Cai, Shenshen Han, Richard Ko, Seth Corrigan, and Kylie A Peppler. 2024. Teachers, Parents, and Students' perspectives on Integrating Generative AI into Elementary Literacy Education. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17.
- [16] Alma Harris and Janet Goodall. 2008. Do parents know they matter? Engaging all parents in learning. *Educational research* 50, 3 (2008), 277–289.
- [17] Hui-Ru Ho, Bengisu Cagiltay, Nathan Thomas White, Edward M Hubbard, and Bilge Mutlu. 2021. RoboMath: Designing a learning companion robot to support children's numerical skills. In *Interaction Design and Children*. 283–293.
- [18] Hui-Ru Ho, Edward M Hubbard, and Bilge Mutlu. 2024. "It's Not a Replacement." Enabling Parent-Robot Collaboration to Support In-Home Learning Experiences of Young Children. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–18.
- [19] Hui-Ru Ho, Nathan White, Edward Hubbard, and Bilge Mutlu. 2023. Designing Parent-child-robot Interactions to Facilitate In-Home Parental Math Talk with Young Children. *arXiv preprint arXiv:2305.02525* (2023).
- [20] Kenneth Holstein, Bruce M McLaren, and Vincent Alevan. 2019. Co-designing a real-time classroom orchestration tool to support teacher-AI complementarity. *Grantee Submission* (2019).
- [21] Kenneth Holstein, Bruce M McLaren, and Vincent Alevan. 2019. Designing for complementarity: Teacher and student needs for orchestration support in AI-enhanced classrooms. In *Artificial Intelligence in Education: 20th International Conference, AIED 2019, Chicago, IL, USA, June 25-29, 2019, Proceedings, Part I* 20. Springer, 157–171.
- [22] Philip N Howard, Lisa-Maria Neudert, Nayana Prakash, and Steven Vosloo. 2021. Digital misinformation/disinformation and children. *UNICEF*. Retrieved on February 20 (2021), 2021.
- [23] Jussi S Jauhainen and Agustin Garagorry Guerra. 2023. Generative AI and ChatGPT in school children's education: Evidence from a school lesson. *Sustainability* 15, 18 (2023), 14025.
- [24] James Kennedy, Paul Baxter, Emmanuel Senft, and Tony Belpaeme. 2016. Social robot tutoring for child second language learning. In *2016 11th ACM/IEEE international conference on human-robot interaction (HRI)*. IEEE, 231–238.
- [25] Christine P Lee, Bengisu Cagiltay, and Bilge Mutlu. 2022. The unboxing experience: Exploration and design of initial interactions between children and social robots. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–14.
- [26] EunKyoung Lee, YoungJun Lee, Bokyoung Kye, and Beomseong Ko. 2008. Elementary and middle school teachers', students' and parents' perception of robot-aided education in Korea. In *EdMedia+ Innovate Learning*. Association for the Advancement of Computing in Education (AACE), 175–183.
- [27] Daniel Leyzberg, Samuel Spaulding, Mariya Toneva, and Brian Scassellati. 2012. The physical presence of a robot tutor increases cognitive learning gains. In *Proceedings of the annual meeting of the cognitive science society*, Vol. 34.
- [28] Edgar Lopez-Caudana, Pedro Ponce, Luis Cervera, Sara Iza, and Nancy Mazon. 2018. Robotic platform for teaching maths in junior high school. *International Journal on Interactive Design and Manufacturing (IJDeM)* 12, 4 (2018), 1349–1360.
- [29] Belinda Louie, Elin A Björling, and Annie Camey Kuo. 2021. The desire for social robots to support English language learners: Exploring robot perceptions of teachers, parents, and students. In *Frontiers in Education*, Vol. 6. Frontiers Media SA, 566909.
- [30] Xin Ma, Jianping Shen, Huilan Y Krenn, Shanshan Hu, and Jing Yuan. 2016. A meta-analysis of the relationship between learning outcomes and parental involvement during early childhood education and early elementary education. *Educational psychology review* 28 (2016), 771–801.
- [31] Joseph E Michaelis and Bilge Mutlu. 2018. Reading socially: Transforming the in-home reading experience with a learning-companion robot. *Science Robotics* 3, 21 (2018), eaat5999.
- [32] Reza Hadi Mogavi, Chao Deng, Justin Juho Kim, Pengyuan Zhou, Young D Kwon, Ahmed Hosny Saleh Metwally, Ahmed Tlili, Simone Bassanelli, Antonio Bucchiarone, Sujit Gujar, et al. 2024. ChatGPT in education: A blessing or a curse? A qualitative study exploring early adopters' utilization and perceptions. *Computers in Human Behavior: Artificial Humans* 2, 1 (2024), 100027.
- [33] Pauline Salim Muljana and Tian Luo. 2021. Utilizing learning analytics in course design: voices from instructional designers in higher education. *Journal of Computing in Higher Education* 33, 1 (2021), 206–234.
- [34] Melina Petsolari, Seray B Ibrahim, and Petr Slovak. 2024. Socio-technical Imaginaries: Envisioning and Understanding AI Parenting Supports through Design Fiction. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–27.
- [35] Aditi Ramachandran, Sarah Strohkorb Sebo, and Brian Scassellati. 2019. Personalized robot tutoring using the assistive tutor POMDP (AT-POMDP). In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 8050–8057.
- [36] Thorsten Schodde, Kirsten Bergmann, and Stefan Kopp. 2017. Adaptive robot language tutoring based on Bayesian knowledge tracing and predictive decision-making. In *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*. 128–136.
- [37] Jiahong Su and Weipeng Yang. 2023. Unlocking the power of ChatGPT: A framework for applying generative AI in education. *ECNU Review of Education* 6, 3 (2023), 355–366.
- [38] Yuling Sun, Jiali Liu, Bingsheng Yao, Jiaju Chen, Dakuo Wang, Xiaojuan Ma, Yuxuan Lu, Ying Xu, and Liang He. 2024. Exploring Parent's Needs for Children-Centered AI to Support Preschoolers' Storytelling and Reading Activities. *arXiv preprint arXiv:2401.13804* (2024).
- [39] Fumihide Tanaka and Takeshi Kimura. 2009. The use of robots in early education: a scenario based on ethical consideration. In *RO-MAN 2009-The 18th IEEE International Symposium on Robot and Human Interactive Communication*. IEEE, 558–560.
- [40] Nils F Tolkdsdorf and Katharina J Rohlfing. 2020. Parents' views on using social robots for language learning. In *2020 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 634–640.
- [41] Kurt VanLehn. 2011. The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational psychologist* 46, 4 (2011), 197–221.
- [42] Kexin Bella Yang, Vanessa Echeverria, Zijing Lu, Hongyu Mao, Kenneth Holstein, Nikol Rummel, and Vincent Alevan. 2023. Pair-up: prototyping human-AI co-orchestration of dynamic transitions between individual and collaborative learning in the classroom. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. 1–17.
- [43] Zhen-Jia You, Chi-Yuh Shen, Chih-Wei Chang, Baw-Jhiune Liu, and Gwo-Dong Chen. 2006. A robot as a teaching assistant in an English class. In *Sixth IEEE international conference on advanced learning technologies (ICALT'06)*. IEEE, 87–91.
- [44] Chao Zhang, Xuechen Liu, Katherine Ziska, Soobin Jeon, Chi-Lin Yu, and Ying Xu. 2024. Mathemyths: leveraging large language models to teach mathematical language through Child-AI co-creative storytelling. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–23.